

Temas de examen

- historia

- Algoritmos: Cesar, vigenère, homofónico, transposición simple, " Atbash, playfair, Hill, Vernam

- Funciones criptográficas (conceptos)

$$A = \{a, b, c, \dots, z\}$$

$$K = \{e_1, e_2, e_3, \dots, e_n\} \quad e \in K$$

$$m = \{m_1, m_2, m_3, \dots, m_n\} \quad m \in M$$

$$\boxed{e, d \in K}$$

$$d = e^{-1}$$

$$E_e(m) = c$$

$$D_d(c) = m$$

$$C = \{c_1, c_2, c_3, \dots, c_n\} \quad c \in C$$

Cifrador Hill

$$E(K, M) = (K \circ M) \bmod 26$$

$m = \text{morning}$

$K = \text{daf}$

$$K \cdot K^{-1} = I \bmod 26$$

$$\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \underbrace{\begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \\ k_{31} & k_{32} & k_{33} \end{bmatrix}}_{[3 \times 3]} \underbrace{\begin{bmatrix} m_1 \\ m_2 \\ m_3 \end{bmatrix}}_{(3 \times 1)} \quad [3 \times 1]$$

$$\begin{cases} c_1 = k_{11}m_1 + k_{12}m_2 + k_{13}m_3 \\ c_2 = k_{21}m_1 + k_{22}m_2 + k_{23}m_3 \\ c_3 = k_{31}m_1 + k_{32}m_2 + k_{33}m_3 \end{cases} \quad \det$$

$$K \neq 0 \bmod 26$$

$$|K| = k_{11}(k_{22}k_{33} - k_{23}k_{32}) - k_{12}(k_{21}k_{33} - k_{23}k_{31}) + k_{13}(k_{21}k_{32} - k_{22}k_{31}) \neq 0 \bmod 26$$

m=morning K=DCDF

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 6 \end{bmatrix}$$

m o r n i n g x
1214 1713 813 623

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 12 \\ 14 \end{bmatrix} = \begin{matrix} 3(12) + 3(14) = 36 + 42 = 78 \mod 26 = 0 & A \\ 2(12) + 5(14) = 24 + 70 = 94 \mod 26 = 16 & Q \end{matrix}$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 17 \\ 13 \end{bmatrix} = \begin{matrix} 12 - M \\ 21 - V \end{matrix}$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 8 \\ 13 \end{bmatrix} = \begin{matrix} 11 - L \\ 3 - D \end{matrix}$$

C = A Q M V L D J X

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 9 \\ 23 \end{bmatrix} = \begin{matrix} 9 - J \\ 23 - X \end{matrix}$$

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 10 & 9 \end{bmatrix} \quad \begin{matrix} A & Q & M & V & L & D & J & X \\ 0 & 16 & 12 & 4 & 11 & 3 & 9 & 23 \end{matrix}$$

↓

$$\begin{bmatrix} 0 \\ 16 \end{bmatrix} = \begin{matrix} 0 + 16(17) = 272 \mod 26 = 12 \\ 0 + 16(9) = 144 \mod 26 = 14 \end{matrix}$$

$$\begin{bmatrix} 12 \\ 21 \end{bmatrix} = \begin{matrix} 12(15) + 21(17) = 537 \mod 26 = 17 \\ 12(20) + 21(9) = 429 \mod 26 = 13 \end{matrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{matrix} 8 & 1 \\ & 13 & 2 \end{matrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} = \begin{matrix} 6 & 6 \\ & 23 & x \end{matrix}$$

Cifrado Vernam

$$A = \{0, 1\}$$

$$K = \{k_1, k_2, k_3, \dots, k_m\}$$

$$m = \{m_1, m_2, m_3, \dots, m_t\}$$

$$E_x = (K_i \oplus m_i) \bmod 2 \quad i = 1, 2, 3, \dots, m$$

$$m = \text{MAR}$$

$$m_1 = m = 12$$

$$K_1 = x = 24$$

$$C_1 = 71 = U$$

$$K = xyz$$

$$\begin{array}{r} \oplus \quad 01100 \\ \quad 10111 \\ \hline 10101 \end{array}$$

MAR
xyz

$$C_1 = U = 21$$

$$K_1 = x = 23$$

$$m_1 = 13 = M$$

$$\begin{array}{r} 16101 \\ 11000 \\ \hline 01101 \end{array}$$